Preliminary HAVOK Results

Hindmarsh-Rose bursting dynamics from scalar observations

AM170B final project snapshot centered on the question: can HAVOK forcing predict burst onset in the Hindmarsh-Rose model?

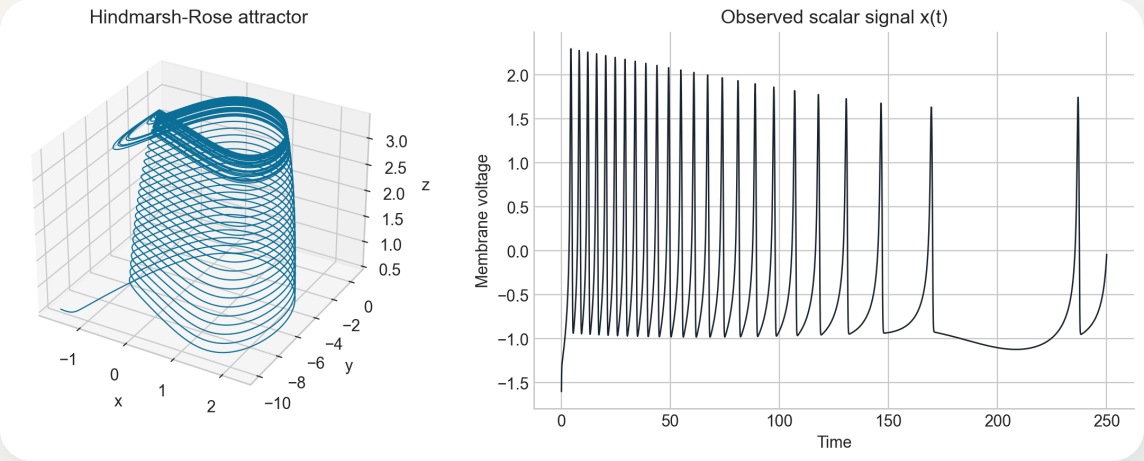
Rank = 11

Delays = 100

100,000 training samples

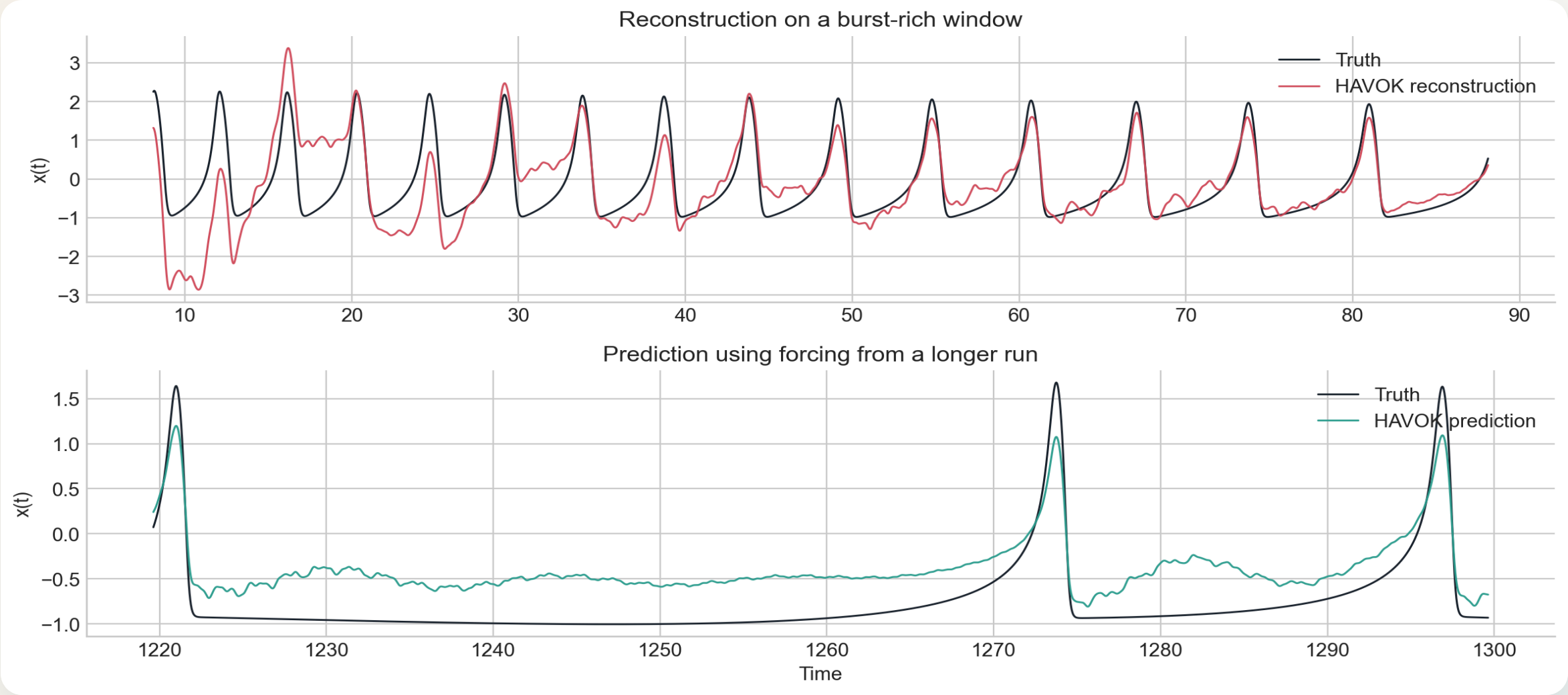
Setup And Framing

- The project replaces the Lorenz tutorial example with the Hindmarsh-Rose neuron model in a chaotic bursting regime.
- Only the scalar voltage-like variable $x(t)$ is supplied to HAVOK; the hidden goal is to recover useful low-dimensional dynamics from delayed coordinates.
- Current script settings use $\Delta t = 0.01$, rank 11, and 100 delay coordinates.
- The extended prediction section now runs after keeping the long forcing model on the same delay setting as the fitted predictor.



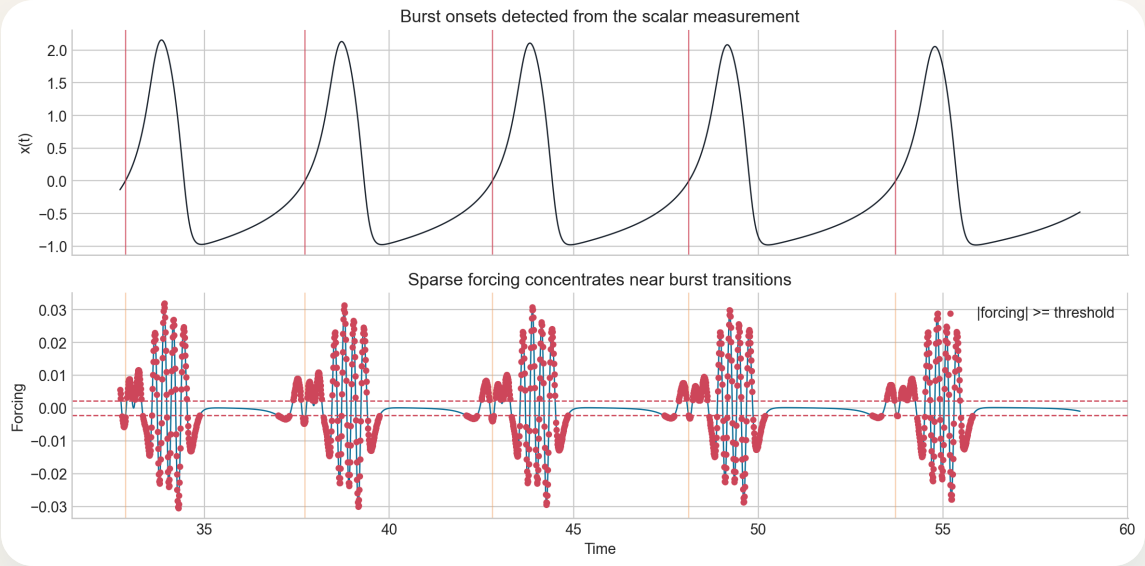
The full Hindmarsh-Rose attractor is generated for reference, but HAVOK is trained only on the scalar trace on the right.

Result 1: Reconstruction Is Stronger Than Forecasting



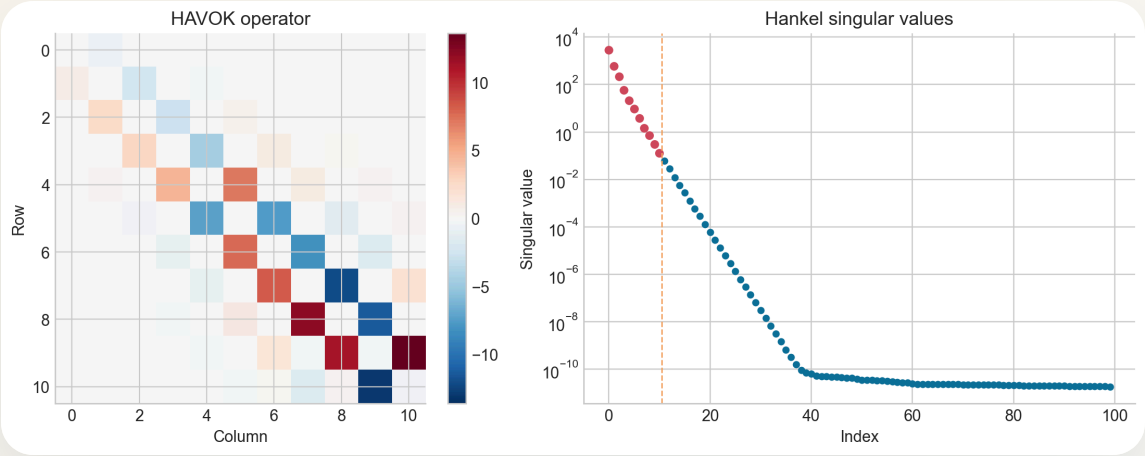
- Reconstruction of the observed training signal is reasonably faithful: correlation ≈ 0.872

Observation / Interpretation



- With the computed threshold, the forcing is active only about **9.7%** of the time.
- Every detected burst onset has a large-forcing event within **0.6 s**, with a median nearest-event gap of about **0.12 s**.
- The important caveat is timing: the nearest active forcing event occurs **before** onset for only about **36%** of bursts, so the signal is usually contemporaneous with onset rather than clearly predictive.
- Interpretation: in this preliminary run, HAVOK forcing behaves more like a transition marker for entry into bursting than a robust early-warning signal.

Result 3: The Embedding Looks Very Low Rank



- The first five singular values already capture about **99.999%** of the Hankel energy; rank 11 is conservative rather than minimal.
- The learned operator is close to the expected skew-symmetric, banded structure, but not perfect: skewness residual $\approx 4.5\%$ and off-tridiagonal mass $\approx 12.2\%$.
- Interpretation: the delayed coordinates are strongly compressible, yet some non-ideal structure remains, which likely contributes to forecast drift.

Expanded Preliminary Results

- From only the scalar signal $x(t)$, HAVOK recovers a compact delayed-coordinate model that reconstructs the training dynamics well: reconstruction correlation ≈ 0.872 with relative RMSE ≈ 0.538.
- The forcing channel is genuinely sparse, active only about **9.7%** of the time, which supports the idea that most of the bursting trajectory is handled by the near-linear delayed subsystem.
- Burst transitions are where the forcing becomes informative: every detected onset has a thresholded forcing event within **0.6 s**, and the median nearest-event gap is only **0.12 s**.
- Longer-run prediction is weaker than reconstruction but still nontrivial: prediction correlation ≈ 0.756, which suggests the model retains qualitative burst structure even when phase accuracy starts to drift.
- The strongest preliminary claim is therefore not full burst-onset prediction yet, but that HAVOK successfully isolates the transition structure of the Hindmarsh-Rose system into a sparse forcing coordinate that stays tightly coupled to bursting events.